

1. Check SELinux status

```
getenforce  
sestatus
```

Meaning:

- getenforce displays the current SELinux mode (Enforcing, Permissive, or Disabled).
- sestatus provides detailed information such as current mode, policy type, and whether SELinux is enabled.

2. Switch to permissive mode

```
sudo setenforce 0  
getenforce
```

```
touch /var/www/html/testfile.txt
```

```
sudo setenforce 1
```

Observation:

In permissive mode, SELinux logs policy violations but does not block the action. In enforcing mode, actions violating SELinux policy may be denied.

3. View and change contexts

```
ls -Z /var/www/html
```

```
sudo chcon --type=httpd_sys_content_t /var/www/html/testfile.txt
```

```
ls -Z /var/www/html/testfile.txt
```

Verify that the SELinux type has changed to httpd_sys_content_t.

4. Analyze denials

```
sudo audit2why < /var/log/audit/audit.log
```

```
sudo audit2allow -a
```

Possible solution:

- Review the denial reason.
- Apply the recommended SELinux policy module if appropriate.
- Ensure the upload directory has the correct SELinux context, for example:

```
sudo chcon -R -t httpd_sys_rw_content_t /srv/foodie/images
```

This context allows the web server to write to the directory when permitted by policy.